

Low-Noise Agile FMCW Source via Self-Injection Locking of a DFB Laser to a Micro-Transfer-Printed TFLT-on-Si₃N₄ Cavity

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Abstract We demonstrate a low-noise, frequency-agile FMCW source by self-injection-locking a DFB laser to a micro-transfer-printed TFLT-on-Si₃N₄ cavity, achieving a 100 Hz linewidth and highly linear frequency sweeps from 1 kHz to 10 MHz. ©2026 The Author(s)

Introduction

Frequency-modulated continuous wave (FMCW) lasers have attracted considerable research attention for high-resolution coherent ranging and imaging systems, with applications spanning light detection and ranging (LiDAR), optical frequency-domain reflectometry (OFDR), and optical coherence tomography (OCT) [1, 2]. Maximum achievable detection range and ranging precision in coherent LiDAR systems are closely related to the laser noise, necessitating laser sources with narrow linewidth to achieve long-distance and high-precision measurements. Moreover, a broad sweep-rate coverage spanning multiple orders of magnitude is essential to accommodate diverse operational scenarios, ranging from high-resolution static imaging to high-speed target tracking. Integrated external cavity lasers, particularly the self-injection-locked lasers (SILs), have emerged as a promising route

for low-noise agile integrated FMCW sources due to their narrow linewidth [3-5]. The chirp performance of the SILs is primarily governed by the modulation response of the external cavity, rendering the selection of the intra-cavity phase tuning mechanism a critical design consideration. The silicon nitride (Si₃N₄) platform enables the fabrication of ultralow-loss waveguides and is therefore highly attractive for narrow-linewidth SILs [6]. However, phase tuning in Si₃N₄ waveguides relies primarily on the thermo-optic effect, which typically offers a limited bandwidth of sub-100 kHz and renders it inadequate for high-speed applications [6]. Although the silicon-on-insulator (SOI) platform enables high-speed phase tuning via the free-carrier plasma-dispersion effect, it introduces additional free-carrier absorption loss that compromises the cavity quality factor (Q-factor) and is therefore detrimental to narrow-linewidth laser operation [7]. Electro-optic (EO)

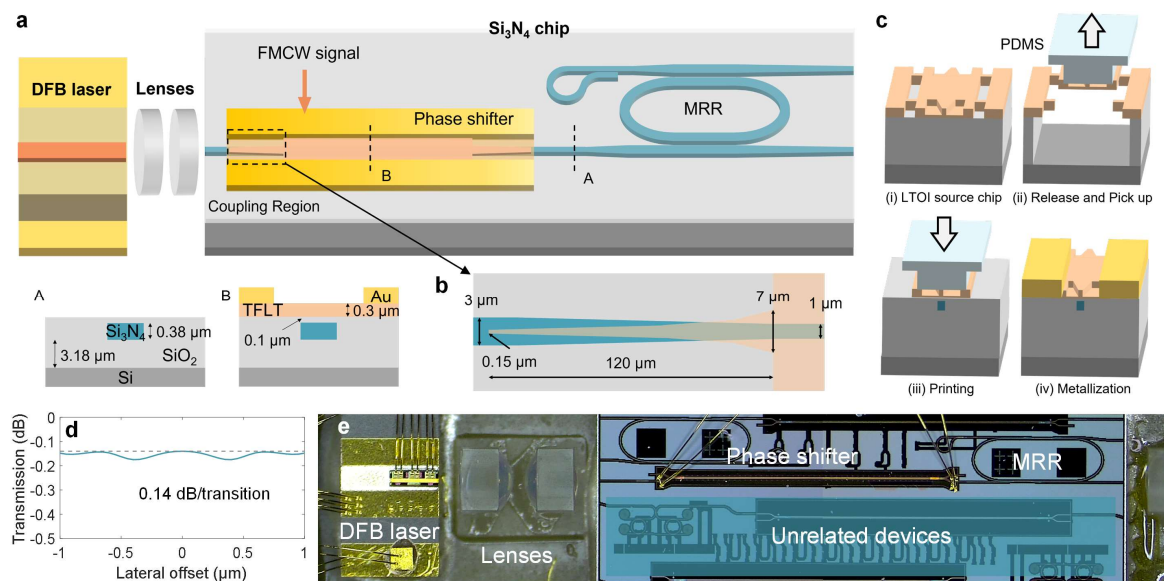


Fig. 1: (a) Schematic of the heterogeneously integrated SIL. (b) Structure of the interlayer coupler. (c) MTP fabrication process. (d) Misalignment tolerance of the interlayer coupler. (e) Microscope image of the packaged laser.

material platforms, such as thin-film lithium niobate (TFLN) and thin-film lithium tantalate (TFLT), offer high linearity and an ultra-broad EO bandwidth surpassing 100 GHz [4, 5, 8, 9], making them suitable for high-speed applications. Nevertheless, TFLN devices are susceptible to direct current (DC) drift effect, which compromises low-frequency stability and long-term chirping accuracy [10]. In addition, the photorefractive effect can degrade the noise performance and stability of TFLN microring-based external cavity lasers. In contrast, TFLT offers markedly enhanced DC stability and significantly reduced photorefractive sensitivity, making it more suitable for high-precision frequency chirping with a wide sweep-rate range [8, 9]. Despite these advantages, TFLT still lags the Si_3N_4 platform in terms of ultralow propagation loss and fabrication maturity. Heterogeneously integrating TFLT with Si_3N_4 , thereby synergistically combining the ultralow-loss performance of Si_3N_4 with the high-speed and highly linear EO tuning capability of TFLT, represents a compelling strategy to overcome these material-level trade-offs and provide a promising pathway toward high-performance lasers.

In this work, we demonstrate an integrated III-V/TFLT/ Si_3N_4 SIL. The III-V distributed feedback (DFB) laser is hybrid-integrated with a Si_3N_4 extension chip that incorporates a heterogeneously integrated TFLT/ Si_3N_4 phase shifter via a

versatile and low-loss micro-transfer-printing (MTP) approach. The SIL exploits the complementary material advantages of InP, TFLT, and Si_3N_4 . By leveraging the high-Q Si_3N_4 microring resonator (MRR), the laser achieves a narrow intrinsic linewidth of 100 Hz. Furthermore, fast and stable tuning of the integrated TFLT/ Si_3N_4 phase shifter enables a wide range of sweep rates from 1 kHz to 10 MHz with high linearity. FMCW fibre-ranging experiments further verify the laser's ranging capability, demonstrating precisions of approximately 0.3 m at 200-m length with 1 kHz sweep rate and 0.07 m at 50-m length with 1 MHz sweep rate.

Laser design and fabrication

Fig. 1(a) presents the schematic of the proposed SIL. The device consists of an InP-based DFB laser butt-coupled to a Si_3N_4 chip that is heterogeneously integrated with a TFLT coupon. The Si_3N_4 chip incorporates a spot size converter, a high-Q MRR, a Sagnac loop reflector, and a hybrid TFLT/ Si_3N_4 EO phase shifter. Since the linewidth suppression ratio of the SIL scales with the square of the MRR Q-factor, the Si_3N_4 waveguide width is designed as 3 μm to minimize scattering loss by reducing the optical field overlap with the sidewalls. To ensure adiabatic transmission of the fundamental mode while maintaining a compact footprint, Euler bends are incorporated into the MRR. Insets A and B in Fig. 1(a)

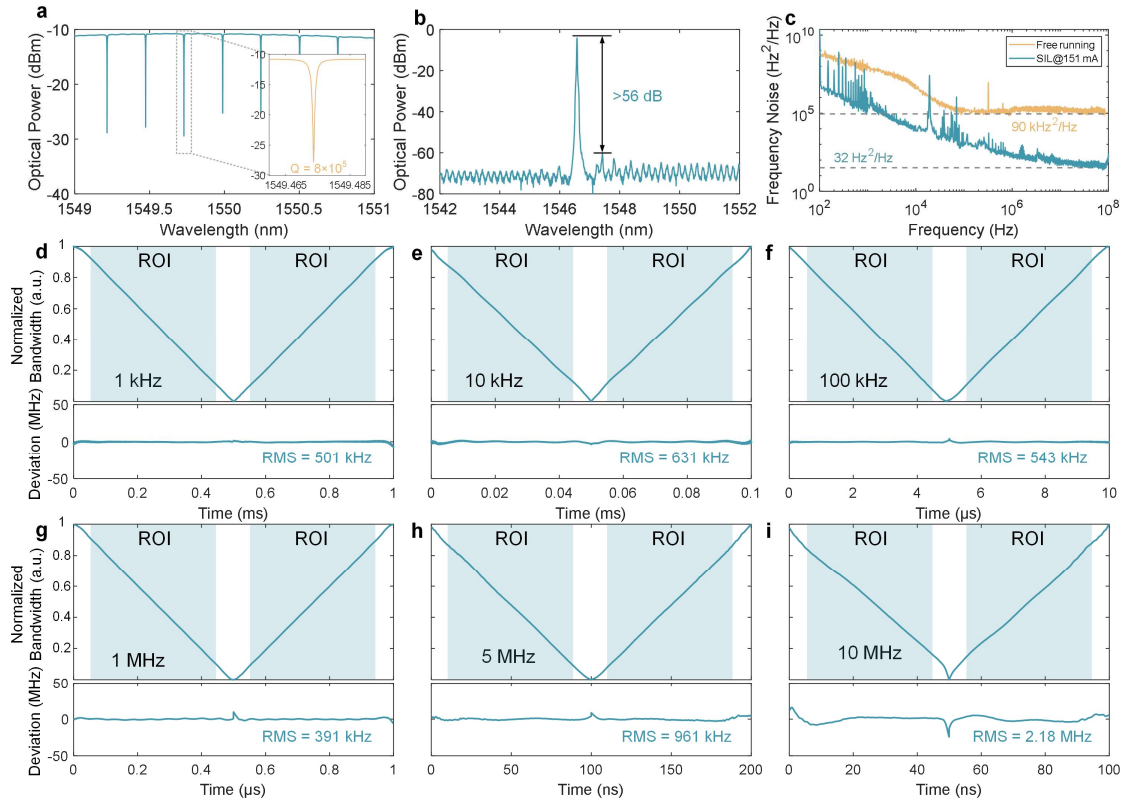


Fig. 2: (a) Transmission spectrum of the Si_3N_4 ring. (b) Lasing spectrum under the pump current of 151 mA. (c) Frequency noise spectra of the laser under the free-running state and the self-injection locking state. (d-i) Top: Time-frequency spectrograms at the modulation speeds from 1 kHz to 10 MHz with 80% range of interest (ROI); Bottom: Residual errors.

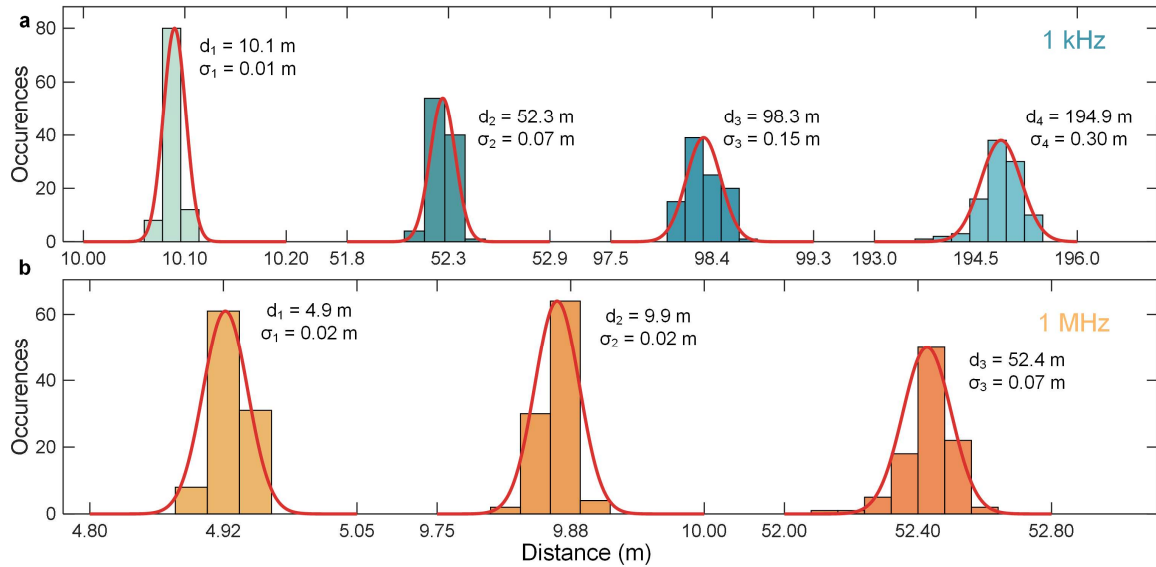


Fig. 3: (a-b) Histogram of measured distance distribution for different fibre lengths at sweep rate of (a) 1 kHz and (b) 1 MHz.

show the cross-sections of the Si_3N_4 waveguide and the TFLT/ Si_3N_4 hybrid waveguide, respectively. A TFLT- Si_3N_4 hybrid waveguide configuration is employed in the phase shifter, enabling higher etching tolerance and minimal propagation loss [11, 12]. In addition, the intrinsically low DC drift of TFLT, combined with the unetched slab design that avoids etch-induced damage, further suppresses DC drift. As shown in Fig. 1(b), a multi-segment interlayer coupler is designed between the Si_3N_4 waveguide and the hybrid waveguide to minimize the transition loss. Detailed optimization methods could be found in our previous works [11, 12]. As shown in Fig. 1(d), the simulated loss remains 0.14 dB to 0.17 dB per transition under lateral offsets above $\pm 1 \mu\text{m}$, indicating both high tolerance and low loss.

Figure 1(c) illustrates the MTP fabrication process. The TFLT coupon, including the unetched region and a pair of interlayer couplers, was first patterned by electron beam lithography and reactive ion etched. After releasing the buried oxide layer by hydrofluoric acid solution, the coupon was picked up using a PDMS stamp, precisely aligned, and printed onto the prepared Si_3N_4 chip. Finally, metal electrodes were deposited using a dual chamber electron beam evaporation coating system (Vikaitech E550D). Figure 1(e) presents the microscope image of the packaged SIL.

Laser performance and FMCW ranging

Figure 2(a) presents the transmission spectrum of the MRR, revealing a free spectral range (FSR) of 32.5 GHz. The inset indicates that the loaded Q-factor reaches 8×10^5 . Figure 2(b) shows the output spectrum of the SIL, demonstrating a side-mode suppression ratio (SMSR) exceeding 56 dB. The optical frequency noise of the SIL is characterized in Fig. 2(c), where the intrinsic linewidth is

extracted to be approximately 100 Hz by multiplying the high-frequency white-noise floor by π . Thanks to the low optical loss of the Si_3N_4 platform, the laser achieves low-noise operation. To generate highly linear FMCW signals, a pre-distortion algorithm was applied at repetition rates of 1 kHz, 10 kHz, 100 kHz, 1 MHz, 5 MHz, and 10 MHz, respectively. As shown in Fig. 2(d-i), the upper panels display time-frequency spectrograms measured with a 5-m-fibre short-delay self-heterodyne setup, indicating chirp linearities of 99.5%-99.92%. The lower panels present the root-mean-square (RMS) deviation of the measured frequency modulation from an ideal triangular ramp. Owing to the excellent EO tuning capability of the hybrid phase shifter, the laser supports a broad sweep-frequency range spanning from 1 kHz to 10 MHz.

Finally, fibre-ranging experiments were performed to evaluate the ranging capability of the SIL, as shown in Fig. 3. Fibre lengths from 10 m to 200 m were measured at a 1 kHz sweep rate, while lengths of 5 m to 50 m were measured at 1 MHz. The system achieved a ranging precision of approximately 0.3 m over 200 m at 1 kHz and 0.07 m over 50 m at 1 MHz.

Conclusions

In this work, we demonstrate a low-noise and frequency-agile FMCW laser realized by self-injection locking a DFB laser to a micro-transfer printed TFLT-on- Si_3N_4 cavity. The hybrid laser achieves a narrow intrinsic linewidth of 100 Hz and enables highly linear frequency sweeping across a broad range from 1 kHz to 10 MHz. In future work, we plan to conduct high-speed and high-precision free-space ranging experiments to further validate the practical feasibility of this FMCW laser.

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